Richness plots

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```
# libraries ----
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.1     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.3     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(readxl)

# data ----

data <- map(1:5, \(x) {
  #read all three taxa
  read_excel("data/Data2024.xlsx", sheet = x) |>
    #pivot
    pivot_longer(-1, names_to = "species") |>
    #rename first column
    rename(plot = 1)
    }
  )

#first three sheets are vascular plants, join them together
data <- bind_rows(data[1:3]) |>
  list() |>
  c(data[4:5])

data <- data |>
  #bind list to single df
  bind_rows(.id = "taxon") |>
  #rename IDs
  mutate(taxon = fct_recode(taxon,
                            "Vascular plants" = "1",
                            "Saproxylic fungi" = "2",
                            "Epiphytic lichens" = "3") |>
```

```r
          fct_relevel("Saproxylic fungi", after = 2),
        #extract genus
        genus = word(species, 1),
        #extract epithets
        epit = word(species, 2),
        #extract site
        site = substr(plot, 1, 2) |>
          fct_recode("Prati di Tivo" = "PR",
                     "Incodaro" = "IN",
                     "Venaquaro" = "VE")
  ) |>
  arrange(taxon, species) |>
  #these were wide matrices, keep only observed entries
  filter(value > 0)

#distinct species names
distinct(data, species) |> pull() |> sort()
```

```
##   [1] "Abies alba"                  "Acer campestre"
##   [3] "Acer opalus"                 "Acer platanoides"
##   [5] "Acer pseudoplatanus"         "Actaea spicata"
##   [7] "Adenostyles australis"       "Adoxa moschatellina"
##   [9] "Aegopodium podagraria"       "Agrimonia eupatoria"
##  [11] "Agrostis capillaris"         "Ajuga reptans"
##  [13] "Allium ursinum"              "Anaptychia ciliaris"
##  [15] "Anaptychia crinalis"         "Aquilegia dumeticola"
##  [17] "Arabis alpina"               "Arabis collina"
##  [19] "Arabis hirsuta"              "Arctium lappa"
##  [21] "Arctium minus"               "Aremonia agrimonoides"
##  [23] "Armillaria sp."              "Arthonia didyma"
##  [25] "Arthonia radiata"            "Arum maculatum"
##  [27] "Asperula laevigata"          "Asplenium ruta-muraria"
##  [29] "Asplenium trichomanes"       "Astragalus glycyphyllos"
##  [31] "Atadinus alpinus"            "Athallia cerinelloides"
##  [33] "Athallia pyracea"            "Atropa bella-donna"
##  [35] "Avenella flexuosa"           "Bertia moriformis"
##  [37] "Biscogniauxia nummularia"    "Bjerkandera adusta"
##  [39] "Brachypodium rupestre"       "Brachypodium sylvaticum"
##  [41] "Bromopsis ramosa"            "Buellia griseovirens"
##  [43] "Bupleurum falcatum"          "Caloplaca cerina"
##  [45] "Caloplaca sp."               "Calycina citrina"
##  [47] "Campanula glomerata"         "Campanula persicifolia"
##  [49] "Campanula rapunculus"        "Campanula tanfanii"
##  [51] "Campanula trachelium"        "Candelariella faginea"
##  [53] "Cardamine bulbifera"         "Cardamine enneaphyllos"
##  [55] "Cardamine graeca"            "Cardamine heptaphylla"
##  [57] "Cardamine kitaibelii"        "Carduus sp"
##  [59] "Carex digitata"              "Carex divulsa"
##  [61] "Carex flacca"                "Carex pilosa"
##  [63] "Carex remota"                "Carex sylvatica"
##  [65] "Centaurium erythraea"        "Cephalanthera damasonium"
##  [67] "Cephalanthera longifolia"    "Cephalanthera rubra"
##  [69] "Cerioporus varius"           "Chlorociboria aeruginascens"
```

```
##  [71] "Chondrostereum purpureum"    "Circaea lutetiana"
##  [73] "Cirsium arvense"             "Clematis vitalba"
##  [75] "Clinopodium menthifolium"    "Clinopodium nepeta"
##  [77] "Clinopodium vulgare"         "Coniophora sp."
##  [79] "Crataegus monogyna"          "Cruciata glabra"
##  [81] "Cruciata laevipes"           "Crustoderma sp."
##  [83] "Cyclamen repandum"           "Cymbalaria muralis"
##  [85] "Cynosurus cristatus"         "Cystopteris fragilis"
##  [87] "Cytisophyllum sessilifolium" "Cytisus scoparius"
##  [89] "Cytisus villosus"            "Dactylis glomerata"
##  [91] "Dactylorhiza maculata"       "Daphne laureola"
##  [93] "Daphne mezereum"             "Dianthus sylvestris"
##  [95] "Diatrype disciformis"        "Diatrype stigma"
##  [97] "Digitalis micrantha"         "Doronicum columnae"
##  [99] "Drymochloa sylvatica"        "Dryopteris filix-mas"
## [101] "Epilobium montanum"          "Epipactis helleborine"
## [103] "Epipactis microphylla"       "Epitrachys italica"
## [105] "Euonymus europaeus"          "Euphorbia amygdaloides"
## [107] "Euphorbia cyparissias"       "Euphrasia stricta"
## [109] "Fagus sylvatica"             "Festuca heterophylla"
## [111] "Fomes fomentarius"           "Fomitopsis pinicola"
## [113] "Fragaria vesca"              "Fraxinus excelsior"
## [115] "Galerina marginata"          "Galium aparine"
## [117] "Galium lucidum"              "Galium odoratum"
## [119] "Galium rotundifolium"        "Ganoderma applanatum"
## [121] "Geranium nodosum"            "Geranium purpureum"
## [123] "Geranium robertianum"        "Glaucomaria carpinea"
## [125] "Grafia golaka"               "Gyalecta sp."
## [127] "Gyalolechia flavorubescens"  "Hedera helix"
## [129] "Helianthemum nummularium"    "Helleborus viridis"
## [131] "Helminthotheca echioides"    "Hepatica nobilis"
## [133] "Heterodermia speciosa"       "Hieracium murorum"
## [135] "Hordelymus europaeus"        "Hypericum montanum"
## [137] "Hypericum perforatum"        "Hypoxylon fragiforme"
## [139] "Hypoxylon sp."               "Ilex aquifolium"
## [141] "Juniperus communis"          "Laburnum anagyroides"
## [143] "Lachnum relicinum"           "Lapsana communis"
## [145] "Laserpitium latifolium"      "Lathyrus pratensis"
## [147] "Lathyrus vernus"             "Lecania naegelii"
## [149] "Lecanora albella"            "Lecanora argentata"
## [151] "Lecanora chlarotera"         "Lecanora intumescens"
## [153] "Lecidella elaeochroma"       "Lenzites betulinus"
## [155] "Lepra amara"                 "Lepraria rigidula"
## [157] "Lepraria sp."                "Lilium bulbiferum"
## [159] "Lilium martagon"             "Limodorum abortivum"
## [161] "Linum viscosum"              "Lonicera alpigena"
## [163] "Lonicera etrusca"            "Lotus corniculatus"
## [165] "Luzula forsteri"             "Luzula multiflora"
## [167] "Luzula sylvatica"            "Medicago lupulina"
## [169] "Megacollybia platyphylla"    "Melanelixia glabratula"
## [171] "Melica uniflora"             "Milium effusum"
## [173] "Moehringia trinervia"        "Monotropa hypophegea"
## [175] "Mycelis muralis"             "Mycena renati"
## [177] "Nemania serpens"             "Neottia nidus-avis"
```

```
## [179] "Ochrolechia pallescens"      "Opegrapha sp."
## [181] "Orthilia secunda"            "Ostrya carpinifolia"
## [183] "Oxalis acetosella"           "Parmelia sulcata"
## [185] "Parmelina spp."              "Pertusaria coccodes"
## [187] "Pertusaria flavida"          "Pertusaria leioplaca"
## [189] "Pertusaria pertusa"          "Petrosedum rupestre"
## [191] "Phaeophyscia insignis"       "Phlyctis agelaea"
## [193] "Phlyctis argena"             "Physcia adscendens"
## [195] "Physcia aipolia"             "Physcia biziana"
## [197] "Physcia leptalea"            "Physcia sp."
## [199] "Physcia stellaris"           "Physconia distorta"
## [201] "Picris hieracioides"         "Platanthera bifolia"
## [203] "Pleurosticta acetabulum"     "Pleurotus cornucopiae"
## [205] "Pluteus cervinus"            "Poa compressa"
## [207] "Poa nemoralis"               "Poa trivialis"
## [209] "Polygonatum odoratum"        "Polyozosia hagenii"
## [211] "Polyporus tuberaster"        "Polystichum aculeatum"
## [213] "Polystichum setiferum"       "Potentilla micrantha"
## [215] "Prenanthes purpurea"         "Primula vulgaris"
## [217] "Prunella vulgaris"           "Prunus avium"
## [219] "Pteridium aquilinum"         "Ptilostemon strictus"
## [221] "Pulmonaria hirta"            "Ramalina farinacea"
## [223] "Ramalina fraxinea"           "Ranunculus lanuginosus"
## [225] "Rosa arvensis"               "Rosa pendulina"
## [227] "Rubus hirtus group"          "Rubus idaeus"
## [229] "Rumex acetosa"               "Ruscus hypoglossum"
## [231] "Salix caprea"                "Sambucus ebulus"
## [233] "Sanicula europaea"           "Saxifraga paniculata"
## [235] "Saxifraga rotundifolia"      "Schizopora paradoxa"
## [237] "Scleroderma verrucosum"      "Scoliciosporum chlorococcum"
## [239] "Scutellinia sp."             "Sedum cepaea"
## [241] "Senecio nemorensis"          "Sesleria autumnalis"
## [243] "Sesleria nitida"             "Silene italica"
## [245] "Solidago virgaurea"          "Sorbus aria"
## [247] "Sorbus aucuparia"            "Stachys sylvatica"
## [249] "Steccherinum robustius"      "Stereum hirsutum"
## [251] "Stereum sp."                 "Stereum subtomentosum"
## [253] "Tanacetum corymbosum"        "Taxus baccata"
## [255] "Tephromela atra"             "Thesium sp"
## [257] "Trametes gibbosa"            "Trametes hirsuta"
## [259] "Trametes ochracea"           "Trametes sp."
## [261] "Trametes versicolor"         "Trifolium campestre"
## [263] "Trifolium ochroleucon"       "Trifolium pratense"
## [265] "Trifolium repens"            "Urtica dioica"
## [267] "Ustulina deusta"             "Verbascum sp"
## [269] "Veronica chamaedrys"         "Veronica cymbalaria"
## [271] "Veronica hederifolia"        "Veronica montana"
## [273] "Veronica officinalis"        "Vicia cracca"
## [275] "Vicia grandiflora"           "Vicia lathyroides"
## [277] "Viola alba"                  "Viola reichenbachiana"
## [279] "Xanthoria parietina"         "Xerocomus subtomentosus"
## [281] "Xylaria carpophila"          "Xylaria hypoxylon"
## [283] "Xylaria longipes"            "Xylaria polymorpha"
## [285] "Xylaria sp."                 "Xylariaceae sp."
```

```r
#entity richness
data |>
  group_by(taxon) |>
  summarise(n_distinct(species), .groups = "drop")
```

```
## # A tibble: 3 x 2
##   taxon              `n_distinct(species)`
##   <fct>                              <int>
## 1 Vascular plants                      191
## 2 Epiphytic lichens                     49
## 3 Saproxylic fungi                      46
```

```r
#distinct genera
distinct(data, genus) |> pull() |> sort()
```

```
##   [1] "Abies"           "Acer"            "Actaea"          "Adenostyles"
##   [5] "Adoxa"           "Aegopodium"      "Agrimonia"       "Agrostis"
##   [9] "Ajuga"           "Allium"          "Anaptychia"      "Aquilegia"
##  [13] "Arabis"          "Arctium"         "Aremonia"        "Armillaria"
##  [17] "Arthonia"        "Arum"            "Asperula"        "Asplenium"
##  [21] "Astragalus"      "Atadinus"        "Athallia"        "Atropa"
##  [25] "Avenella"        "Bertia"          "Biscogniauxia"   "Bjerkandera"
##  [29] "Brachypodium"    "Bromopsis"       "Buellia"         "Bupleurum"
##  [33] "Caloplaca"       "Calycina"        "Campanula"       "Candelariella"
##  [37] "Cardamine"       "Carduus"         "Carex"           "Centaurium"
##  [41] "Cephalanthera"   "Cerioporus"      "Chlorociboria"   "Chondrostereum"
##  [45] "Circaea"         "Cirsium"         "Clematis"        "Clinopodium"
##  [49] "Coniophora"      "Crataegus"       "Cruciata"        "Crustoderma"
##  [53] "Cyclamen"        "Cymbalaria"      "Cynosurus"       "Cystopteris"
##  [57] "Cytisophyllum"   "Cytisus"         "Dactylis"        "Dactylorhiza"
##  [61] "Daphne"          "Dianthus"        "Diatrype"        "Digitalis"
##  [65] "Doronicum"       "Drymochloa"      "Dryopteris"      "Epilobium"
##  [69] "Epipactis"       "Epitrachys"      "Euonymus"        "Euphorbia"
##  [73] "Euphrasia"       "Fagus"           "Festuca"         "Fomes"
##  [77] "Fomitopsis"      "Fragaria"        "Fraxinus"        "Galerina"
##  [81] "Galium"          "Ganoderma"       "Geranium"        "Glaucomaria"
##  [85] "Grafia"          "Gyalecta"        "Gyalolechia"     "Hedera"
##  [89] "Helianthemum"    "Helleborus"      "Helminthotheca"  "Hepatica"
##  [93] "Heterodermia"    "Hieracium"       "Hordelymus"      "Hypericum"
##  [97] "Hypoxylon"       "Ilex"            "Juniperus"       "Laburnum"
## [101] "Lachnum"         "Lapsana"         "Laserpitium"     "Lathyrus"
## [105] "Lecania"         "Lecanora"        "Lecidella"       "Lenzites"
## [109] "Lepra"           "Lepraria"        "Lilium"          "Limodorum"
## [113] "Linum"           "Lonicera"        "Lotus"           "Luzula"
## [117] "Medicago"        "Megacollybia"    "Melanelixia"     "Melica"
## [121] "Milium"          "Moehringia"      "Monotropa"       "Mycelis"
## [125] "Mycena"          "Nemania"         "Neottia"         "Ochrolechia"
## [129] "Opegrapha"       "Orthilia"        "Ostrya"          "Oxalis"
## [133] "Parmelia"        "Parmelina"       "Pertusaria"      "Petrosedum"
## [137] "Phaeophyscia"    "Phlyctis"        "Physcia"         "Physconia"
## [141] "Picris"          "Platanthera"     "Pleurosticta"    "Pleurotus"
## [145] "Pluteus"         "Poa"             "Polygonatum"     "Polyozosia"
## [149] "Polyporus"       "Polystichum"     "Potentilla"      "Prenanthes"
```

```
## [153] "Primula"        "Prunella"       "Prunus"         "Pteridium"
## [157] "Ptilostemon"    "Pulmonaria"     "Ramalina"       "Ranunculus"
## [161] "Rosa"           "Rubus"          "Rumex"          "Ruscus"
## [165] "Salix"          "Sambucus"       "Sanicula"       "Saxifraga"
## [169] "Schizopora"     "Scleroderma"    "Scoliciosporum" "Scutellinia"
## [173] "Sedum"          "Senecio"        "Sesleria"       "Silene"
## [177] "Solidago"       "Sorbus"         "Stachys"        "Steccherinum"
## [181] "Stereum"        "Tanacetum"      "Taxus"          "Tephromela"
## [185] "Thesium"        "Trametes"       "Trifolium"      "Urtica"
## [189] "Ustulina"       "Verbascum"      "Veronica"       "Vicia"
## [193] "Viola"          "Xanthoria"      "Xerocomus"      "Xylaria"
## [197] "Xylariaceae"
```

```r
#genus richness
data |>
  #Xylariaceae is not a genus, don't count
  filter_out(genus == "Xylariaceae") |>
  group_by(taxon) |>
  summarise(n_distinct(genus), .groups = "drop")
```

```
## # A tibble: 3 x 2
##   taxon              `n_distinct(genus)`
##   <fct>                            <int>
## 1 Vascular plants                    132
## 2 Epiphytic lichens                   31
## 3 Saproxylic fungi                    33
```

```r
#distinct epithets
distinct(data, epit) |> pull() |> sort()
```

```
##   [1] "abortivum"       "acetabulum"      "acetosa"         "acetosella"
##   [5] "aculeatum"       "adscendens"      "adusta"          "aeruginascens"
##   [9] "agelaea"         "agrimonoides"    "aipolia"         "alba"
##  [13] "albella"         "alpigena"        "alpina"          "alpinus"
##  [17] "amara"           "amygdaloides"    "anagyroides"     "aparine"
##  [21] "applanatum"      "aquifolium"      "aquilinum"       "argena"
##  [25] "argentata"       "aria"            "arvense"         "arvensis"
##  [29] "atra"            "aucuparia"       "australis"       "autumnalis"
##  [33] "avium"           "baccata"         "bella-donna"     "betulinus"
##  [37] "bifolia"         "biziana"         "bulbifera"       "bulbiferum"
##  [41] "campestre"       "capillaris"      "caprea"          "carpinea"
##  [45] "carpinifolia"    "carpophila"      "cepaea"          "cerina"
##  [49] "cerinelloides"   "cervinus"        "chamaedrys"      "chlarotera"
##  [53] "chlorococcum"    "ciliaris"        "citrina"         "coccodes"
##  [57] "collina"         "columnae"        "communis"        "compressa"
##  [61] "corniculatus"    "cornucopiae"     "corymbosum"      "cracca"
##  [65] "crinalis"        "cristatus"       "cymbalaria"      "cyparissias"
##  [69] "damasonium"      "deusta"          "didyma"          "digitata"
##  [73] "dioica"          "disciformis"     "distorta"        "divulsa"
##  [77] "dumeticola"      "ebulus"          "echioides"       "effusum"
##  [81] "elaeochroma"     "enneaphyllos"    "erythraea"       "etrusca"
##  [85] "eupatoria"       "europaea"        "europaeus"       "excelsior"
##  [89] "faginea"         "falcatum"        "farinacea"       "filix-mas"
```

```
##  [93] "flacca"          "flavida"         "flavorubescens"  "flexuosa"
##  [97] "fomentarius"     "forsteri"        "fragiforme"      "fragilis"
## [101] "fraxinea"        "gibbosa"         "glabra"          "glabratula"
## [105] "glomerata"       "glycyphyllos"    "golaka"          "graeca"
## [109] "grandiflora"     "griseovirens"    "hagenii"         "hederifolia"
## [113] "helix"           "helleborine"     "heptaphylla"     "heterophylla"
## [117] "hieracioides"    "hirsuta"         "hirsutum"        "hirta"
## [121] "hirtus"          "hypoglossum"     "hypophegea"      "hypoxylon"
## [125] "idaeus"          "insignis"        "intumescens"     "italica"
## [129] "kitaibelii"      "laevigata"       "laevipes"        "lanuginosus"
## [133] "lappa"           "lathyroides"     "latifolium"      "laureola"
## [137] "leioplaca"       "leptalea"        "longifolia"      "longipes"
## [141] "lucidum"         "lupulina"        "lutetiana"       "maculata"
## [145] "maculatum"       "marginata"       "martagon"        "menthifolium"
## [149] "mezereum"        "micrantha"       "microphylla"     "minus"
## [153] "monogyna"        "montana"         "montanum"        "moriformis"
## [157] "moschatellina"   "multiflora"      "muralis"         "murorum"
## [161] "naegelii"        "nemoralis"       "nemorensis"      "nepeta"
## [165] "nidus-avis"      "nitida"          "nobilis"         "nodosum"
## [169] "nummularia"      "nummularium"     "ochracea"        "ochroleucon"
## [173] "odoratum"        "officinalis"     "opalus"          "pallescens"
## [177] "paniculata"      "paradoxa"        "parietina"       "pendulina"
## [181] "perforatum"      "persicifolia"    "pertusa"         "pilosa"
## [185] "pinicola"        "platanoides"     "platyphylla"     "podagraria"
## [189] "polymorpha"      "pratense"        "pratensis"       "pseudoplatanus"
## [193] "purpurea"        "purpureum"       "pyracea"         "radiata"
## [197] "ramosa"          "rapunculus"      "reichenbachiana" "relicinum"
## [201] "remota"          "renati"          "repandum"        "repens"
## [205] "reptans"         "rigidula"        "robertianum"     "robustius"
## [209] "rotundifolia"    "rotundifolium"   "rubra"           "rupestre"
## [213] "ruta-muraria"    "scoparius"       "secunda"         "serpens"
## [217] "sessilifolium"   "setiferum"       "sp"              "sp."
## [221] "speciosa"        "spicata"         "spp."            "stellaris"
## [225] "stigma"          "stricta"         "strictus"        "subtomentosum"
## [229] "subtomentosus"   "sulcata"         "sylvatica"       "sylvaticum"
## [233] "sylvestris"      "tanfanii"        "trachelium"      "trichomanes"
## [237] "trinervia"       "trivialis"       "tuberaster"      "uniflora"
## [241] "ursinum"         "varius"          "vernus"          "verrucosum"
## [245] "versicolor"      "vesca"           "villosus"        "virgaurea"
## [249] "viridis"         "viscosum"        "vitalba"         "vulgare"
## [253] "vulgaris"
```

```r
#check incerta
incerta_index <- c("", "sp.", "spp.")

incerta <- data |>
  filter(epit %in% incerta_index) |>
  distinct(taxon, species, genus)

incerta
```

```
## # A tibble: 15 x 3
##    taxon              species         genus
##    <fct>              <chr>           <chr>
```

```
##  1 Epiphytic lichens Caloplaca sp.   Caloplaca
##  2 Epiphytic lichens Gyalecta sp.    Gyalecta
##  3 Epiphytic lichens Lepraria sp.    Lepraria
##  4 Epiphytic lichens Opegrapha sp.   Opegrapha
##  5 Epiphytic lichens Parmelina spp.  Parmelina
##  6 Epiphytic lichens Physcia sp.     Physcia
##  7 Saproxylic fungi  Armillaria sp.  Armillaria
##  8 Saproxylic fungi  Coniophora sp.  Coniophora
##  9 Saproxylic fungi  Crustoderma sp. Crustoderma
## 10 Saproxylic fungi  Hypoxylon sp.   Hypoxylon
## 11 Saproxylic fungi  Scutellinia sp. Scutellinia
## 12 Saproxylic fungi  Stereum sp.     Stereum
## 13 Saproxylic fungi  Trametes sp.    Trametes
## 14 Saproxylic fungi  Xylaria sp.     Xylaria
## 15 Saproxylic fungi  Xylariaceae sp. Xylariaceae
```

```r
#check congenerics of incerta

data |>
  filter(genus %in% incerta$genus) |>
  distinct(species, taxon) |>
  print(n = 50)
```

```
## # A tibble: 33 x 2
##    species               taxon
##    <chr>                 <fct>
##  1 Caloplaca cerina      Epiphytic lichens
##  2 Caloplaca sp.         Epiphytic lichens
##  3 Gyalecta sp.          Epiphytic lichens
##  4 Lepraria rigidula     Epiphytic lichens
##  5 Lepraria sp.          Epiphytic lichens
##  6 Opegrapha sp.         Epiphytic lichens
##  7 Parmelina spp.        Epiphytic lichens
##  8 Physcia adscendens    Epiphytic lichens
##  9 Physcia aipolia       Epiphytic lichens
## 10 Physcia biziana       Epiphytic lichens
## 11 Physcia leptalea      Epiphytic lichens
## 12 Physcia sp.           Epiphytic lichens
## 13 Physcia stellaris     Epiphytic lichens
## 14 Armillaria sp.        Saproxylic fungi
## 15 Coniophora sp.        Saproxylic fungi
## 16 Crustoderma sp.       Saproxylic fungi
## 17 Hypoxylon fragiforme  Saproxylic fungi
## 18 Hypoxylon sp.         Saproxylic fungi
## 19 Scutellinia sp.       Saproxylic fungi
## 20 Stereum hirsutum      Saproxylic fungi
## 21 Stereum sp.           Saproxylic fungi
## 22 Stereum subtomentosum Saproxylic fungi
## 23 Trametes gibbosa      Saproxylic fungi
## 24 Trametes hirsuta      Saproxylic fungi
## 25 Trametes ochracea     Saproxylic fungi
## 26 Trametes sp.          Saproxylic fungi
## 27 Trametes versicolor   Saproxylic fungi
## 28 Xylaria carpophila    Saproxylic fungi
```

```
## 29 Xylaria hypoxylon     Saproxylic fungi
## 30 Xylaria longipes      Saproxylic fungi
## 31 Xylaria polymorpha    Saproxylic fungi
## 32 Xylaria sp.           Saproxylic fungi
## 33 Xylariaceae sp.       Saproxylic fungi
```

```r
#those that have no congenerics

keep_genus <- data |>
  filter(genus %in% incerta$genus) |>
  distinct(genus, species, taxon) |>
  add_count(genus) |>
  filter(n == 1)

keep_genus
```

```
## # A tibble: 8 x 4
##   genus       species          taxon                 n
##   <chr>       <chr>            <fct>             <int>
## 1 Gyalecta    Gyalecta sp.     Epiphytic lichens     1
## 2 Opegrapha   Opegrapha sp.    Epiphytic lichens     1
## 3 Parmelina   Parmelina spp.   Epiphytic lichens     1
## 4 Armillaria  Armillaria sp.   Saproxylic fungi      1
## 5 Coniophora  Coniophora sp.   Saproxylic fungi      1
## 6 Crustoderma Crustoderma sp.  Saproxylic fungi      1
## 7 Scutellinia Scutellinia sp.  Saproxylic fungi      1
## 8 Xylariaceae Xylariaceae sp.  Saproxylic fungi      1
```

These should be kept because they are the only ones in their genus. Now, let's check the ones with more entries:

```r
data |>
  filter(genus %in% incerta$genus) |>
  distinct(genus, species, taxon) |>
  add_count(genus) |>
  filter(n > 1) |>
  print(n = 50)
```

```
## # A tibble: 25 x 4
##    genus     species                taxon                 n
##    <chr>     <chr>                  <fct>             <int>
##  1 Caloplaca Caloplaca cerina       Epiphytic lichens     2
##  2 Caloplaca Caloplaca sp.          Epiphytic lichens     2
##  3 Lepraria  Lepraria rigidula      Epiphytic lichens     2
##  4 Lepraria  Lepraria sp.           Epiphytic lichens     2
##  5 Physcia   Physcia adscendens     Epiphytic lichens     6
##  6 Physcia   Physcia aipolia        Epiphytic lichens     6
##  7 Physcia   Physcia biziana        Epiphytic lichens     6
##  8 Physcia   Physcia leptalea       Epiphytic lichens     6
##  9 Physcia   Physcia sp.            Epiphytic lichens     6
## 10 Physcia   Physcia stellaris      Epiphytic lichens     6
## 11 Hypoxylon Hypoxylon fragiforme   Saproxylic fungi      2
## 12 Hypoxylon Hypoxylon sp.          Saproxylic fungi      2
```

```
## 13 Stereum   Stereum hirsutum      Saproxylic fungi      3
## 14 Stereum   Stereum sp.           Saproxylic fungi      3
## 15 Stereum   Stereum subtomentosum Saproxylic fungi      3
## 16 Trametes  Trametes gibbosa      Saproxylic fungi      5
## 17 Trametes  Trametes hirsuta      Saproxylic fungi      5
## 18 Trametes  Trametes ochracea     Saproxylic fungi      5
## 19 Trametes  Trametes sp.          Saproxylic fungi      5
## 20 Trametes  Trametes versicolor   Saproxylic fungi      5
## 21 Xylaria   Xylaria carpophila    Saproxylic fungi      5
## 22 Xylaria   Xylaria hypoxylon     Saproxylic fungi      5
## 23 Xylaria   Xylaria longipes      Saproxylic fungi      5
## 24 Xylaria   Xylaria polymorpha    Saproxylic fungi      5
## 25 Xylaria   Xylaria sp.           Saproxylic fungi      5
```

Of those, *Lepraria* is kept because it was different from *Lepraria rigidula*, while *Physcia* and *Caloplaca* should be removed.

```
keep_genus <- c(keep_genus$genus, "Lepraria")

#keep only those incerta
incerta <- incerta |>
  filter(!(genus %in% keep_genus))

data <- data |>
  anti_join(incerta)
```

```
## Joining with ‘by = join_by(taxon, species, genus)‘
```

```
# mean richness per area ---------------------------------------------------

data |>
  summarise(richness = n_distinct(species), .by = c(taxon, site, plot)) |>
  summarise(avg = mean(richness), .by =c(taxon, site)) |>
  mutate(avg = round(avg))
```

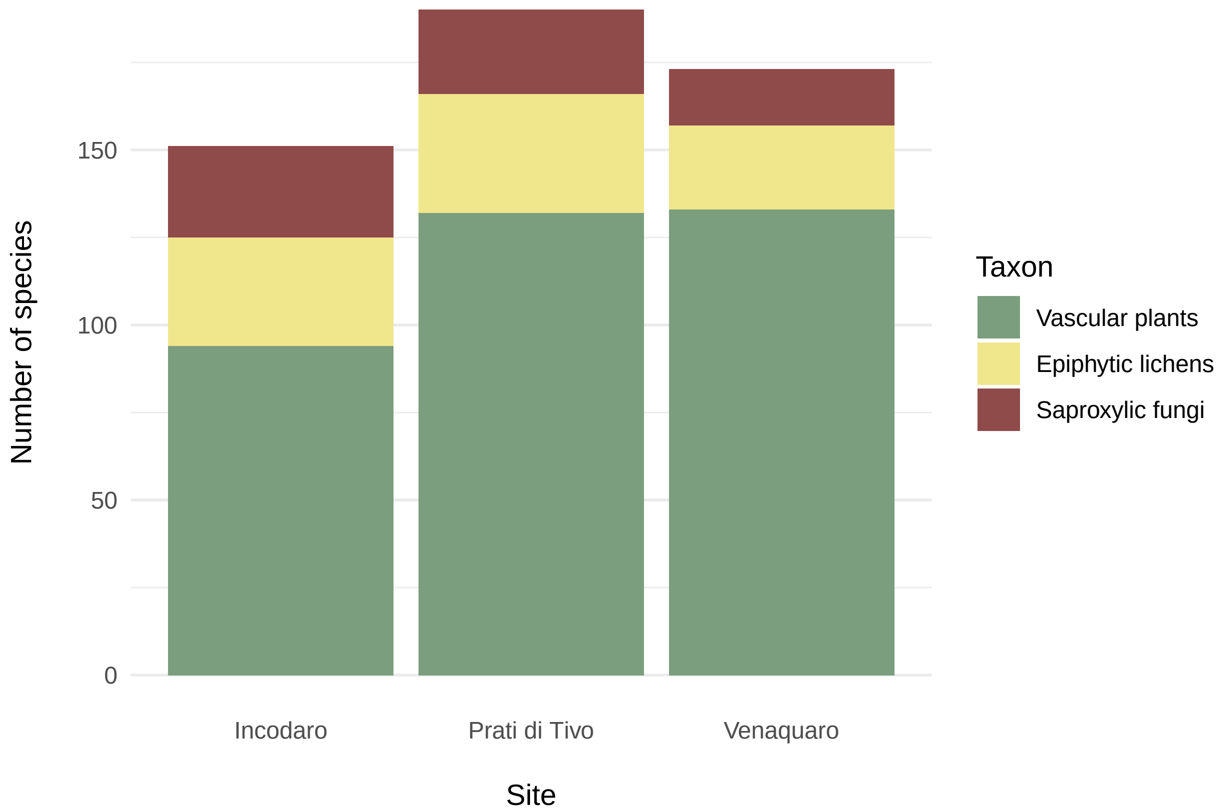
```
## # A tibble: 9 x 3
##   taxon              site             avg
##   <fct>              <fct>          <dbl>
## 1 Vascular plants   Incodaro          31
## 2 Vascular plants   Venaquaro         56
## 3 Vascular plants   Prati di Tivo     66
## 4 Epiphytic lichens Prati di Tivo     16
## 5 Epiphytic lichens Incodaro          14
## 6 Epiphytic lichens Venaquaro         12
## 7 Saproxylic fungi  Venaquaro          4
## 8 Saproxylic fungi  Incodaro           5
## 9 Saproxylic fungi  Prati di Tivo      7
```

```
# plot! --------------------------------------------------------------------

richness <- data |>
  group_by(site, taxon) |>
```

```r
  summarise(rich = n_distinct(species), .groups = "drop")

richness |>
  arrange(taxon, desc(rich))
```

```
## # A tibble: 9 x 3
##   site          taxon              rich
##   <fct>         <fct>             <int>
## 1 Venaquaro     Vascular plants     133
## 2 Prati di Tivo Vascular plants     132
## 3 Incodaro      Vascular plants      94
## 4 Prati di Tivo Epiphytic lichens    34
## 5 Incodaro      Epiphytic lichens    31
## 6 Venaquaro     Epiphytic lichens    24
## 7 Incodaro      Saproxylic fungi     26
## 8 Prati di Tivo Saproxylic fungi     24
## 9 Venaquaro     Saproxylic fungi     16
```

```r
richness |>
  ggplot(aes(x = site, y = rich, fill = taxon)) +
  geom_col(position = position_stack(reverse = TRUE)) +
  scale_fill_discrete(palette = c("#7B9E7E", "#F0E68C", "#8F4A4A")) +
  theme_minimal() +
  labs(
    x = "Site",
    y = "Number of species",
    fill = "Taxon"
  ) +
  theme(
    axis.title.x = element_text(margin = margin(t = 15)),
    axis.title.y = element_text(margin = margin(r = 15)),
    plot.title = element_text(hjust = 0.5, size = 16, face = "bold", margin = margin(b = 15)),
    panel.grid.major.x = element_blank()
  )
```




```
ggsave("richness_stack.png", width = 119, height = 73, scale = 1.2, units = "mm", bg = "white")
ggsave("Fig2.eps", width = 119, height = 73, scale = 1.2, units = "mm", bg = "white")
```